Singular Value Decomposition

January 12, 2026 8:32 AM

Consider matrix A which is $m \times n$

The right singular vectors/values of A:

$$\underbrace{A^T A \vec{v} = \lambda \vec{v}}$$

The left singular vectors/values

$$A A^T \vec{u} = \sigma \vec{u}$$

$$A A^T (A \vec{v}) = \lambda (A \vec{v})$$

$A\vec{v}$ is an eigenvectors of AA^T.

$$A\vec{v} \propto \vec{u}$$

$$|A\vec{v}| = \sqrt{(\vec{v} A)^T (A\vec{v})}$$

$$= \sqrt{\vec{v}^T A^T A \vec{v}}$$

$$= \sqrt{\vec{v}^T \lambda \vec{v}}$$

$$= \sqrt{\lambda}$$

$$\frac{A\vec{v}}{\sqrt{\lambda}} = \vec{u}$$

$$A\vec{v} = \vec{u}\sqrt{\lambda}$$

This is true for all $\vec{v}_i$ and $\vec{u}_i$

$$A\vec{v}_i = \vec{u}_i\sqrt{\lambda_i} \quad \forall i$$

$$AV = UD$$

V is matrix whose columns are right singular vectors of A

U is matrix whose columns are left singular vectors of A

D is a diagonal whose elements are roots of singular values.

$$A = UDV^T$$

This is SVD!

Example

Compute SVD of

$$A = \begin{bmatrix} -1 & 0 & -2 \\ 0 & 1 & 2 \end{bmatrix}$$

Right singular vectors:

$$A^TA = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 2 & 2 & 8 \end{bmatrix}$$

$$\lambda_1 = 0, \quad \vec{v}_1 = \begin{bmatrix} 2/3 \\ 2/3 \\ -1/3 \end{bmatrix}$$

$$\lambda_2 = 1, \quad \vec{v}_2 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$\lambda_3 = 9, \quad \vec{v}_3 = \begin{bmatrix} 1/\sqrt{18} \\ 1/\sqrt{18} \\ 4/\sqrt{18} \end{bmatrix}$$

Left singular vectors:

$$AA^T = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}$$

$$\lambda_1 = 1, \quad \vec{u}_1 = \begin{bmatrix} -1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

$$\lambda_2 = 9, \quad \vec{u}_2 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$V = \begin{bmatrix} 2/3 & 1/\sqrt{2} & 1/\sqrt{18} \\ 2/3 & -1/\sqrt{2} & 1/\sqrt{18} \\ & & 4/ \end{bmatrix} \qquad U = \begin{bmatrix} -1/\sqrt{2} & -1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$\begin{bmatrix} 2/3 & -1/\sqrt{2} & 1/\sqrt{18} \\ -1/3 & 0 & 4/\sqrt{18} \end{bmatrix} \qquad \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

UDV^T

D has 3 columns

D has 2 rows

$$D = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$A = UDV^T$$

$$= \begin{bmatrix} -1/\sqrt{2} & -1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 2/3 & 2/3 & -1/3 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{18} & 1/\sqrt{18} & 4/\sqrt{18} \end{bmatrix}$$

Model of Biological Neurons

Dendrites - receive info from adjacent neurons

Axons - send info to adjacent neurons

Axons -

Body - processes info and controls activities

Synapse - connection between neurons

Types of synapses:

1. Excitatory - synapses where higher signal leads to higher response

2. Inhibitory - synapses where higher signal leads to lower response